



# Study on the Co-gasification characteristics of biomass and municipal solid waste based on machine learning

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## ABSTRACT

Co-gasification of biomass and municipal solid waste (MSW) exhibits synergistic effects by improving the quality of syngas while reducing environmental pollution from MSW. In this study, Machine learning (ML) techniques were employed to investigate the co-gasification process of biomass and MSW. A comprehensive dataset was constructed using existing data, including different feedstock types and operating conditions, with 18 input features and 9 output features. Four advanced ML models were utilized to model and analyze the co-gasification process. By leveraging feedstock characteristics and operating parameters, key gasification parameters such as syngas composition, lower heating value (LHV) of syngas, tar yield, and carbon conversion efficiency were predicted. The results showed that all four models exhibited excellent predictive performance, with  $R^2$  values greater than 0.9 in both the training and testing stage. Specifically, Histogram-based gradient boosting regression (HGBR) exhibited the lowest root mean square error (RMSE) in predicting CO, while the gradient boosting regressor (GBR) achieved the best performance in  $H_2$  prediction with a RMSE of 1.6. The most influential input features for CO concentration were equivalence ratio (ER), oxygen content in biomass and hydrogen content in biomass. The key features affecting  $H_2$  concentration were steam/fuel and ER.

## 1. Introduction

In recent years, rapid population growth and increasing industrialization across countries have led to significant energy demands, raising concerns about the depletion of fossil resources [1]. The extensive use of fossil fuels results in high emissions of greenhouse gases such as  $CO_2$  and  $CH_4$ , further endangering the natural environment on which human survival depends [2]. To address this issue, seeking sustainable energy development has become a global topic of common interest [3]. Currently, research is focused on finding alternatives to fossil fuels, such as renewable energy sources like solar power, wind energy, and biomass energy. Biomass energy, as a typical carbon-neutral energy source, has advantages over solar and wind energy in terms of being unrestricted by time and geographical limitations. Compared to fossil fuels, biomass energy also has a smaller environmental impact and can be used

worldwide [4]. Biomass resources are abundant globally, and according to forecasts by the International Energy Agency, biomass resources could provide a sustainable supply to meet 27 % of the global energy consumption by 2050 [5]. Based on the aforementioned characteristics of biomass, the efficient utilization of biomass resources remains a key area of research on a global scale.

Currently, there are various methods for the utilization of biomass resources, such as thermochemical conversion, biochemical conversion, and chemical conversion [6]. Thermochemical conversion plays a crucial role in biomass utilization due to its high conversion efficiency, zero waste nature, and favorable economic characteristics. Common thermochemical conversion methods include combustion, gasification, and pyrolysis [7]. Gasification and pyrolysis refer to the thermal decomposition of biomass in the temperature range of 523–1673 K under oxygen-deficient or oxygen-free conditions. Through gasification and pyrolysis process, high-quality syngas, valuable fuels, or certain

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**Abbreviations**

|                               |                                                |
|-------------------------------|------------------------------------------------|
| W                             | Blending ratio of MSW to biomass               |
| C <sub>b</sub>                | The carbon content in biomass                  |
| H <sub>b</sub>                | The hydrogen content in biomass                |
| O <sub>b</sub>                | The oxygen content in biomass                  |
| N <sub>b</sub>                | The nitrogen in biomass                        |
| V <sub>b</sub>                | The volatile matter in biomass                 |
| FC <sub>b</sub>               | The fixed carbon in biomass                    |
| Ash <sub>b</sub>              | The ash content in biomass                     |
| ER                            | Equivalence ratio                              |
| Stram/fuel                    | Steam to feedstock ratio                       |
| C <sub>n</sub> H <sub>m</sub> | Hydrocarbons of C <sub>2</sub> ~C <sub>4</sub> |
| XGBR                          | Extreme gradient boosting regression           |
| PCA                           | Principal component analysis                   |
| CCE                           | Carbon conversion efficiency                   |
| MAE                           | Mean absolute error                            |

|                  |                                              |
|------------------|----------------------------------------------|
| LHV              | Lower heating value of syngas                |
| MSW              | Municipal solid waste                        |
| C <sub>m</sub>   | The carbon content in MSW                    |
| H <sub>m</sub>   | The hydrogen content in MSW                  |
| O <sub>m</sub>   | The oxygen content in MSW                    |
| N <sub>m</sub>   | The nitrogen in MSW                          |
| V <sub>m</sub>   | The volatile matter in MSW                   |
| FC <sub>m</sub>  | The fixed carbon in MSW                      |
| Ash <sub>m</sub> | The ash content in MSW                       |
| SHAP             | SHapley Additive exPlanations                |
| GBR              | Gradient boosting regressor                  |
| RFR              | Random forest regression                     |
| HGBR             | Histogram-based gradient boosting regression |
| RMSE             | Root mean square error                       |
| R <sup>2</sup>   | Coefficient of determination                 |
| ML               | Machine learning                             |

chemical raw materials can be obtained from biomass. Compared to pyrolysis, gasification produces cleaner gases, resulting in higher energy recovery efficiency and being more advantageous in terms of heat capacity [8]. Currently, extensive research has been conducted on biomass gasification, including various gasification reactors, the performance of different types of biomass in gasification, and the impact of operational parameters on the gasification process [9–11].

The quality of syngas produced through biomass gasification depends largely on the biomass feedstock used. Due to existing challenges in biomass gasification, such as seasonality, high oxygen content leading to lower heating value, corrosion issues, and instability resulting from varying compositions and lower pyrolysis rates [11,12]. Some researchers have started investigating the co-gasification of MSW and biomass to overcome the limitations of solely using biomass as a feedstock [13]. MSW mainly consists of paper, food waste, plastics, rubber, textiles, wood, glass, and sludge, among others. Organic compounds make up more than 80 % of MSW and can be used as a source of energy. Global urban MSW generation is projected to exceed 2.2 billion tons by 2025, a 40 % increase from 2012 [14]. A synergetic effect was reported upon the co-gasification of biomass and MSW: the former has a lower stability, it can easily decompose and generate volatiles which promote the degradation of polymers in the MSW at lower temperatures, thus resulting in an improvement of the properties and quality of the final product, since more hydrogen is produced in the syngas [15]. Additionally, co-gasification facilitates the utilization of MSW as a valuable resource, avoiding environmental impacts. Therefore, the co-gasification of biomass and MSW is a win-win process.

The co-gasification process of biomass and MSW is a complex thermochemical conversion process, influenced by various factors such as the composition of biomass and MSW, as well as operating parameters (temperature, equivalence ratio, gasifying agent, etc.). Currently, many researchers have investigated the co-gasification process of biomass and MSW, primarily through experimental means [16]. Experimental approaches generally yield localized conclusions, and a significant number of biomass and MSW types and operating conditions often require a large number of experimental trials to optimize the input variables for industrial applications. However, this approach can be time-consuming and resource-intensive [17]. In addition to experimental methods, process simulation and computational fluid dynamics are mainstream research tools. However, due to the complex heat and mass transfer characteristics of the co-gasification process, these methods have not been widely used in the industrial field of co-gasification [18].

It is gratifying that with the rapid advancement of hardware and software capabilities, the application of ML in thermochemical conversion reactions is becoming increasingly common, overcoming some of

the limitations mentioned earlier [19]. ML models are data-driven and can handle high-dimensional, complex, and nonlinear data, offering advantages such as low cost and short processing time [20]. This modeling approach is more effective and efficient in handling large and complex co-gasification datasets compared to traditional statistical methods. ML models, based on end-to-end learning, can bypass the need to understand the complex nonlinear chemical reactions in the gasification process. Instead, they focus on establishing the input-output relationships to gain profound insights into the gasification process and assist in industrial design [21]. Currently, ML has been widely used in the gasification process. Detailed information on the current state of research on gasification using ML is provided in Table 1.

The existing literature has demonstrated the applicability of ML methods in gasification. Currently, there is a considerable amount of research on the individual thermal decomposition of biomass, but there is limited literature specifically focused on co-gasification processes. Yang et al. [23] studied the co-gasification of coal and biomass; however, their study had limited features, with output only including gas yield and LHV of gas, which cannot provide a comprehensive explanation of gasification performance. Particularly, it does not address the reutilization of components in syngas, such as hydrogen production, therefore lacking reference value. Based on this, for the first time, this study employs ML methods to investigate the co-gasification process of biomass and MSW. Ample input and output features (18 input features, 9 output features) were selected to build a database encompassing different gasification conditions and various types of biomass and MSW. Four advanced ML models (Extreme gradient boosting regression (XGBR), Gradient boosting regression (GBR), Random forest regression (RFR), Histogram-based gradient boosting regression (HGBR)) were utilized to model the co-gasification process and evaluate the predictive performance of the models. SHAP (SHapley Additive exPlanations) analysis based on game theory was employed to interpret the models, aiming to clarify the influence of biomass and MSW composition and operating conditions on gasification performance parameters. Importantly, steam-assisted gasification features were also incorporated in this study to provide guidance for co-gasification hydrogen production.

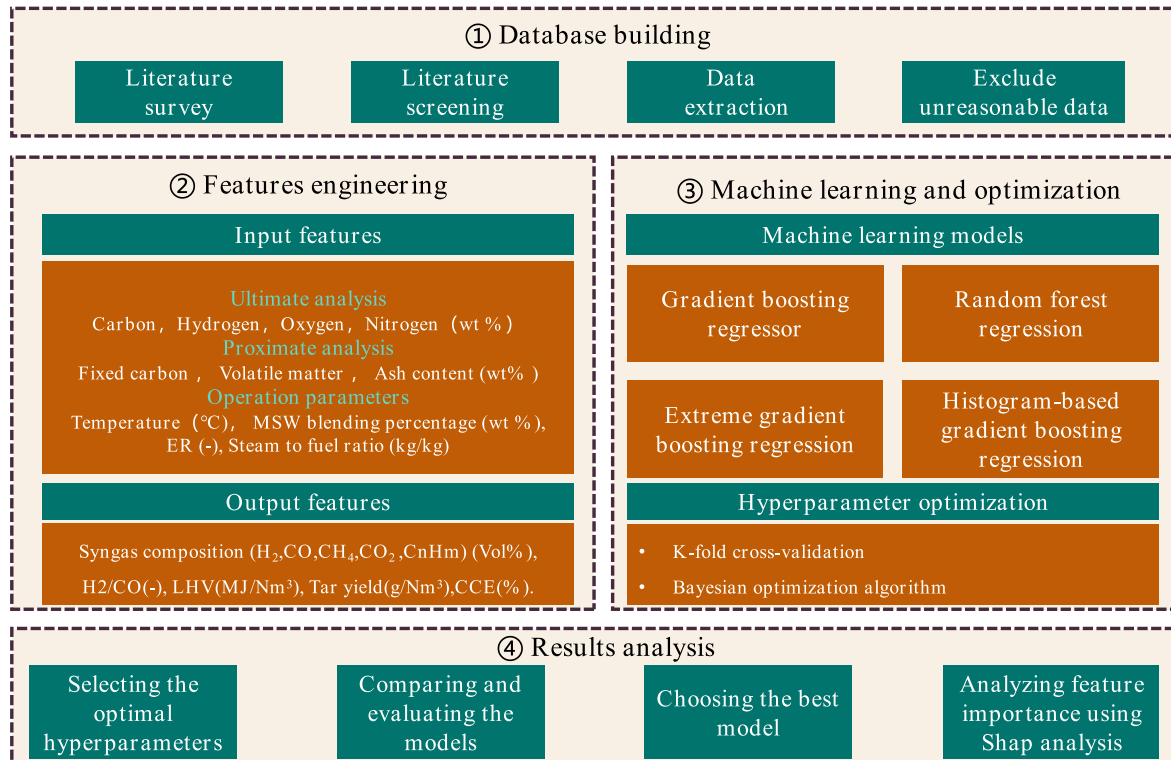
## 2. Methodology

This study aims to use ML to study the complex thermochemical conversion process of biomass and municipal solid waste (MSW) co-gasification, in order to facilitate the production of high-quality combustible gas. The research content and workflow of this article are shown in Fig. 1. First, data collection and feature engineering were performed. By conducting searches with relevant keywords in existing

**Table 1**

A literature review of research on ML models in gasification.

| Ref. | Feedstock type    | ML models                                                                                                       | Reactor type                                                        | Gasifying agent       | Input                                                                                                                                                                                     | Output                                                                                                                                                                   |
|------|-------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [22] | Biomass           | Nearest neighbor, decision tree, random forest, extreme gradient boosting.                                      | Bubbling bed, pressurized fluidized bed, circulating fluidized bed. | Air or steam.         | Ultimate analysis (C, H, O), proximate analysis (ash content, fixed carbon, volatile matter), operation parameters (temperature, steam to biomass ratio, ER).                             | Syngas composition (H <sub>2</sub> , CO, CO <sub>2</sub> , CH <sub>4</sub> , H <sub>2</sub> /CO)                                                                         |
| [23] | Coal and biomass  | Decision tree regression, artificial neural network, support vector machine.                                    | Fluidized bed;                                                      | Air                   | Ultimate analysis (C, H, O), proximate analysis (ash content, fixed carbon, volatile matter), operation parameters (temperature).                                                         | Gas yield, LHV of gas.                                                                                                                                                   |
| [17] | MSW               | Support vector machine, random forest, gradient boost regressor                                                 | Fixed bed, fluidized bed.                                           | Air                   | Ultimate analysis (C, H, O, N, S), proximate analysis (ash content, fixed carbon, volatile matter), operation parameters (temperature, ER).                                               | Syngas composition (H <sub>2</sub> , CO, CO <sub>2</sub> , CH <sub>4</sub> )<br>syngas yield, LHV, char yield, tar yield.                                                |
| [24] | Biomass           | AutoML framework                                                                                                | Bubbling bed, pressurized fluidized bed, circulating fluidized bed. | Air or steam.         | Syngas composition (H <sub>2</sub> , CO, CO <sub>2</sub> , CH <sub>4</sub> ), LHV, char yield, tar yield.                                                                                 | Biomass composition, ER, temperature, pressure, steam to biomass ratio, superficial gas velocity.                                                                        |
| [25] | Biomass and waste | Artificial neural network                                                                                       | Fixed bed, Fluidized bed.                                           | Air, steam or oxygen. | Ultimate analysis (C, H, S), proximate analysis (ash, moisture), operation parameters (particle size, operation mode, temperature, ER, catalyst, gasifying agent, gasifier bed material). | Syngas composition (N <sub>2</sub> , H <sub>2</sub> , CO, CO <sub>2</sub> , C <sub>2</sub> H <sub>6</sub> , CH <sub>4</sub> ), LHV, char yield, syngas yield, tar yield. |
| [26] | Biomass and waste | Random forest, gradient, boosting for regression, multilayer perceptron neural network, support vector machine. | Fixed bed, fluidized bed.                                           | Air, steam or oxygen. | Ultimate analysis (C, H, S), proximate analysis (ash, moisture), operation parameters (particle size, operation mode, temperature, ER, catalyst, gasifying agent, gasifier bed material). | Syngas composition (N <sub>2</sub> , H <sub>2</sub> , CO, CO <sub>2</sub> , C <sub>2</sub> H <sub>6</sub> , CH <sub>4</sub> ), LHV, char yield, syngas yield, tar yield. |
| [27] | Biomass           | Artificial neural network, random forest, support vector machine.                                               | Fluidized bed.                                                      | Air, steam.           | Lignocellulose composition, temperature, pressure, equivalence ratio, steam to biomass ratio, superficial gas velocity.                                                                   | Syngas composition (H <sub>2</sub> , CO, CO <sub>2</sub> , CH <sub>4</sub> ), LHV, char yield, tar yield.                                                                |

**Fig. 1.** The research content and process employed in this research.

databases, we obtained relevant data and manually organized it to meet the requirements of ML. The selected input features include the ultimate analysis and proximate analysis of biomass and MSW, as well as the operational parameters of gasification. The output features consist of syngas composition,  $H_2/CO$  ratio, LHV of syngas, tar yield, and carbon conversion efficiency (CCE), among others. These output features cover important performance parameters of the co-gasification process. Once the appropriate features are selected, advanced ML models are used to model the co-gasification process of biomass and MSW. The selected ML models primarily include tree-based models such as XGBR, GBR, RFR, and HGBR. These models have been proven to perform exceptionally well when modeling structured data compared to neural network models [28]. During the training process of these models, K-fold cross-validation is used to evaluate the models, enhancing their generalization performance and minimizing overfitting. Additionally, Bayesian optimization algorithms are employed to fine-tune hyperparameters to obtain the optimal hyperparameters for each model. After the training is completed, quantitative metrics are used to evaluate the models, ultimately selecting the best predictive model. Based on the optimal model, SHAP analysis is conducted to assess the importance of input features in relation to their impact on the corresponding output.

## 2.1. Data extraction and preprocessing

The required data for the research was retrieved from the Scopus database. The search was conducted using keywords such as “Co-gasification,” “MSW,” “Biomass,” “Plastic,” etc., resulting in approximately 857 articles. After careful screening, articles that did not meet the criteria were excluded, resulting in a final set of 255 valid data samples (Supplementary Excel File). The collected MSW types include plastics, rubber, tires, waste paper, etc., and biomass types include forestry and agricultural biomass. The input features of the collected data include the ultimate analysis (carbon (wt%), hydrogen (wt%), nitrogen (wt%), oxygen (wt%)) and proximate analysis (ash content (wt%), fixed carbon (wt%), volatile matter (wt%)) of biomass and MSW, as well as operational parameters (temperature ( $^{\circ}C$ ), MSW blending ratio (wt%), ER (–), and steam/fuel ratio (kg/kg)). The output responses include the composition of syngas ( $H_2$ , CO,  $CO_2$ ,  $CH_4$ ,  $C_nH_m$ ) (vol%),  $H_2/CO$  ratio, syngas LHV value ( $MJ/Nm^3$ ), tar yield ( $g/Nm^3$ ), and CCE (%).

In general, a significant amount of valid data is required for effective ML models. The amount of data needed depends on the research question, selected model, and their relationship. Some literature suggests that the required data number should be around 5 to 10 times the number of independent input parameters [17]. In this study, the data collected for tar yield and CCE-related parameters is relatively limited. However, it is still possible to train a ML model with these parameters while preventing overfitting. For missing values in the input parameters, they can be supplemented by using the corresponding category's average values.

After obtaining reasonable input values, it is necessary to analyze the relationships between various features. Pyrolysis and gasification are complex chemical processes, and the relationships between the features are difficult to describe using linear correlations. Therefore, in this study, Spearman correlation coefficient analysis was employed to assess the correlations between all features. This method employs rank transformations to determine the strength and direction of the correlation by using monotonic functions [29]. The core idea is to perform rank transformations (RX and RY) for two variables separately and then calculate the correlation between RX and RY following the methodology of Pearson correlation analysis. If there are no duplicate values in the data and the two variables are perfectly monotonic correlated, the Spearman correlation coefficient would be +1 or –1. A value of +1 indicates a perfect positive correlation, –1 indicates a perfect negative correlation, and other results fall between –1 and +1. The expression for Spearman correlation coefficient  $r_s$  is as follows :

$$r_s = 1 - \frac{6 \sum d_i^2}{N(N^2-1)} \quad (1)$$

where ,  $d_i$  is the difference between each pair of the ranked variables and  $N$  is the overall number of samples.

When there are a large number of input features, dimensionality reduction analysis is an effective approach to understand the relationships between these features. Principal component analysis (PCA) aims to transform multiple features into a smaller number of comprehensive features, known as principal components, using the idea of dimensionality reduction. It is a linear transformation that maps the data into a new coordinate system, where the first principal component captures the largest variance of the data projection, the second principal component captures the second largest variance, and so on. PCA is often used to reduce the dimensionality of a dataset while retaining the features that contribute the most to the dataset's variance. This is achieved by preserving the lower-order principal components and ignoring the higher-order ones. By doing this, the lower-order components can retain the most important aspects of the data. This article employs PCA to analyze the interrelationships between the key features during the co-gasification process of biomass and MSW. Additionally, a contour plot is used to study the impact of operational parameters on the four major gas composition ( $H_2$ , CO,  $CH_4$ , and  $CO_2$ ).

## 2.2. ML model development

In this study, a random splitting method is used, where 80 % of the data is allocated to the training set for training and validation during the model training process, while the remaining data is treated as the test set to evaluate the model's performance. Four ML models, namely GBR, XGBR, RFR, and HGBR, are employed. These models have been proven to have good adaptability in pyrolysis and gasification processes [21]. GBR is a Boosting algorithm that iteratively constructs multiple decision tree models, with each model attempting to correct the errors of the previous one. It is based on the concept of gradient boosting, minimizing the gradient of the loss function to build the trees. GBR exhibits good generalization and robustness in scenarios with structured data and relatively simple feature engineering, making it capable of handling non-linear relationships and high-dimensional features. XGBR is an improved version of GBR, introducing regularization, enhanced performance, and more customization options. It also includes regularization terms in the objective function to prevent overfitting. RFR is a Bagging algorithm that builds multiple decision tree models in parallel and then aggregates their results for predictions. Each decision tree is trained on a different random subset of data, increasing model diversity. The prediction result of RFR is the average of multiple decision trees. HGBR aims to provide an efficient implementation of Gradient Boosting Trees. HGBR uses histograms to speed up the training process. Unlike GBR, which splits the feature space all at once, HGBR first bins the features into discrete histograms and then constructs decision trees on each histogram. This approach reduces the complexity of splitting the feature space and enhances training speed.

Cross-validation is an effective technique to prevent overfitting during the model training process. In this study, five-fold cross-validation is employed, where the training set is divided into five equal parts, with four parts used for training and one part used for validation in each iteration. This process is repeated until all data points have been used for model validation. This study is conducted using the scikit-learn library, a ML library developed in the Python programming language.

In order to evaluate the performance of each model, this study utilizes several metrics, including the  $R^2$ , RMSE, and MAE. The calculations for these metrics are as follows:

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (2)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (3)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (4)$$

where  $N$  represents the number of samples,  $y_i$ ,  $\hat{y}_i$  and  $\bar{y}$  represent the actual values, the predicted values, and the mean of the experimental values, respectively.

Hyperparameter optimization is an important part of the training process for models. In this study, the Optuna open-source library is used to perform hyperparameter optimization for the various ML models. Optuna automates the process of finding the best hyperparameters for ML models, primarily using Bayesian optimization and random search techniques. Optuna intelligently selects hyperparameter combinations to evaluate, aiming to minimize or maximize the objective function, in order to efficiently find the optimal combination within the search space of hyperparameters.

### 2.3. SHAP feature analysis

It is well known that ML models are often considered as black-box models, making it challenging to interpret their behavior. SHAP analysis, based on game theory, provides a method to explain the impact of each input feature on the model's output response. It achieves this by considering various permutations and combinations of feature values to determine the contribution of each feature to the output response. SHAP analysis offers a fair way to allocate changes in predicted values to each individual input feature in the model. It quantifies the contribution of each feature to the model's predictions. This approach allows for the calculation of feature importance, detection of feature interactions, and provides insights into global model behavior, among other information. Currently, many researchers have utilized SHAP analysis to interpret ML models used for modeling processes such as pyrolysis and gasification [30].

## 3. Results and discussion

### 3.1. Data feature analysis

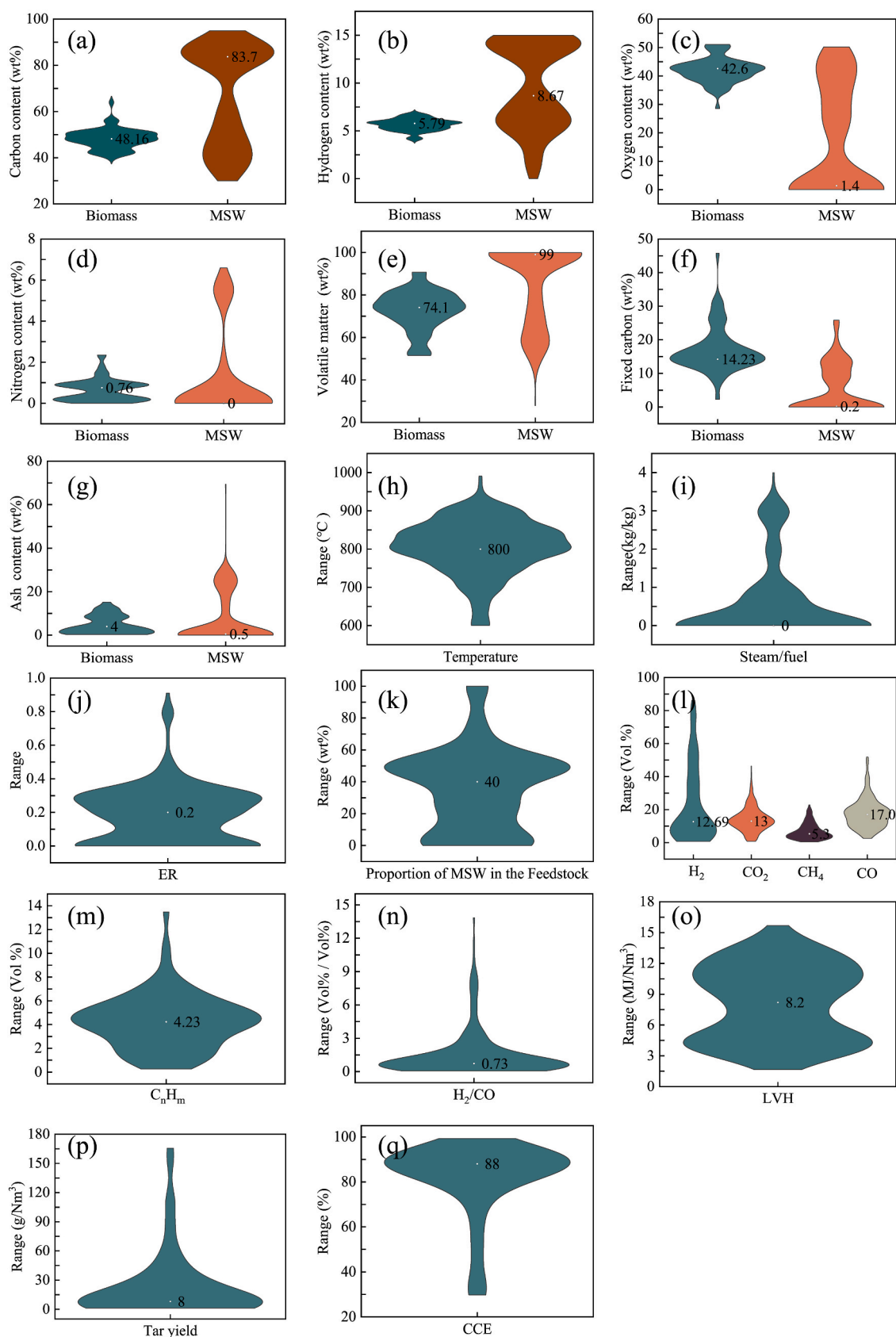
Through statistical analysis, it is possible to gain a better understanding of the data distribution of various features in biomass and MSW. This is essential for building an effective ML model and ensuring its reliability. The statistical distributions of each feature can be seen in Fig. 2. From the graph, one can observe the distribution of each feature, including the range of minimum and maximum values, median values, and the concentration of samples. Carbon content is the most abundant element in biomass and MSW, with ranges of 38 %–68 % and 30 %–90 %, respectively. The broader range of carbon content in MSW is primarily due to its diverse composition. High molecular weight polymers such as plastics and waste tires contain higher carbon content, while materials like waste paper have lower carbon content. Adding MSW with high carbon content to biomass during co-pyrolysis processes increases the feasibility of producing carbon-rich fuels through this process. There are also significant differences in hydrogen content between biomass and MSW. The hydrogen content ranges from 3.5 % to 7 % for biomass and from 0.5 % to 15 % for MSW, with median values of 5.79 % and 8.67 %, respectively. The addition of MSW with high hydrogen content in co-gasification processes is beneficial as it promotes the production of  $H_2$  to a large extent. On the other hand, hydrogen supply can facilitate the decomposition of organic substances during co-gasification processes [31]. The median oxygen content for biomass and MSW is 42.6 % and 1.4 %, respectively. Low oxygen content has a certain inhibitory effect on the generation of oxygen-containing substances such as alcohols, ketones, and aldehydes in co-gasification processes. In comparison to biomass, MSW has higher volatile matter and lower fixed carbon

content, which contributes to the production of combustible gases during gasification. Volatile matter undergoes thermal cracking at high temperatures to generate small molecules of combustible gas [32]. At the same time, the lower fixed carbon content requires less oxygen for its conversion to CO.

In terms of operating parameters, temperature and ER are the most crucial gasification operating parameters, fluctuating in the ranges of 600 to 1000 °C and 0 to 0.9, respectively. These parameters encompass typical gasification conditions, including slight oxygen deficiency pyrolysis and approximate direct combustion. A wider range of parameter variations can effectively improve the model's generalization performance. Steam-assisted gasification is also an important direction in gasification, and it is included in this study. The Steam/fuel input feature represents the amount of steam added and can be used to study its impact on the output features. Its variation range is between 0 and 4. The MSW blending ratio represents the amount of MSW added, ranging from 0 % to 100 %, with a median value of 40 %. This indicates that a significant amount of MSW can be consumed in the co-gasification process, allowing for its resource utilization.

Regarding the output features, the main syngas composition, including the volume fractions of  $CO$ ,  $H_2$ ,  $CH_4$ , and  $CO_2$ , has median values of 17.0 %, 12.69 %, 5.3 %, and 13.0 %, respectively. These values also fluctuate over a significant range, particularly for  $H_2$ , which fluctuates between 3 % and 80 %. This variation can be attributed to the feedstock characteristics and significant variations in operating parameters. Additionally, the introduction of steam as a gasifying agent is a key factor in increasing the hydrogen content.  $H_2/CO$  is also included in this study to assess the extent of hydrogen generation, with a median value of 0.73. Furthermore, the LHV, tar yield, and CCE studied, with median values of 8.2 MJ/Nm<sup>3</sup>, 8 g/Nm<sup>3</sup>, and 88 %, respectively. The study of these output features provides a more comprehensive understanding of the mechanisms involved in the co-gasification process. In summary, the distribution ranges of all the mentioned features are reasonably balanced, neither excessively concentrated nor excessively dispersed. This meets the data requirements of ML models.

Fig. 3 depicts the Spearman correlation coefficients among the different features. It can be observed that the concentration of  $H_2$  has a strong positive correlation with Steam/fuel, while it exhibits a strong negative correlation with ER, with correlation coefficients of 0.84 and  $-0.76$ , respectively. The addition of steam is a commonly used method in gasification processes to generate hydrogen. Through steam reforming reactions, hydrocarbons can be converted into hydrogen, promoting the water-gas shift reaction and methane reforming reaction, thereby enhancing hydrogen production [33]. An increase in ER leads to a higher oxygen content in the reactor, resulting in the production of more greenhouse gases instead of  $H_2$  in the oxidizing zone of the reactor [11]. The same pattern can be observed in the correlation between the features  $H_2/CO$  and ER, with a correlation coefficient of  $-0.67$ . The concentration of  $CO$  is positively correlated with the oxygen content in biomass (0.43), primarily because some of the oxygen for  $CO$  production comes from the feedstock itself. Temperature is the most crucial parameter in gasification processes. It exhibits a certain positive correlation with the volume fractions of  $CO$  and  $H_2$  (0.14 and 0.09, respectively). High temperatures facilitate the further decomposition of organic matter, resulting in the production of more  $CO$  and  $H_2$  and inhibiting the formation of hydrocarbons, mainly attributed to the acceleration of Boudouard and water-gas shift reactions [34]. Temperature increases the value of CCE (0.51) but decreases the LHV of syngas ( $-0.13$ ), indicating that high-temperature gasification suppresses gasification performance. However, high temperatures accelerate the tar decomposition process ( $-0.48$ ), leading to a reduction in tar production. Steam/fuel appears to have a positive correlation with  $CO$ ,  $H_2$ , and  $CH_4$ , indicating that the addition of steam greatly facilitates the conversion of carbon and hydrogen into combustible gas. The blending ratio of MSW shows a good positive correlation with  $C_nH_m$  and the LHV of syngas, with correlation coefficients of 0.24 and 0.39, respectively. This is



**Fig. 2.** Statistical distribution of collected data. (a) Carbon content, (b) Hydrogen content, (c) Oxygen content, (d) Nitrogen content, (e) Volatile matter, (f) Fixed carbon content, (g) Ash content, (h) Temperature, (i) Steam to fuels ratio, (j) ER, (k) MSW blending ratio, (l) Syngas composition, (m) C<sub>n</sub>H<sub>m</sub> composition, (n) H<sub>2</sub>/CO, (o) LHV, (p) Tar yield, (q) CCE.

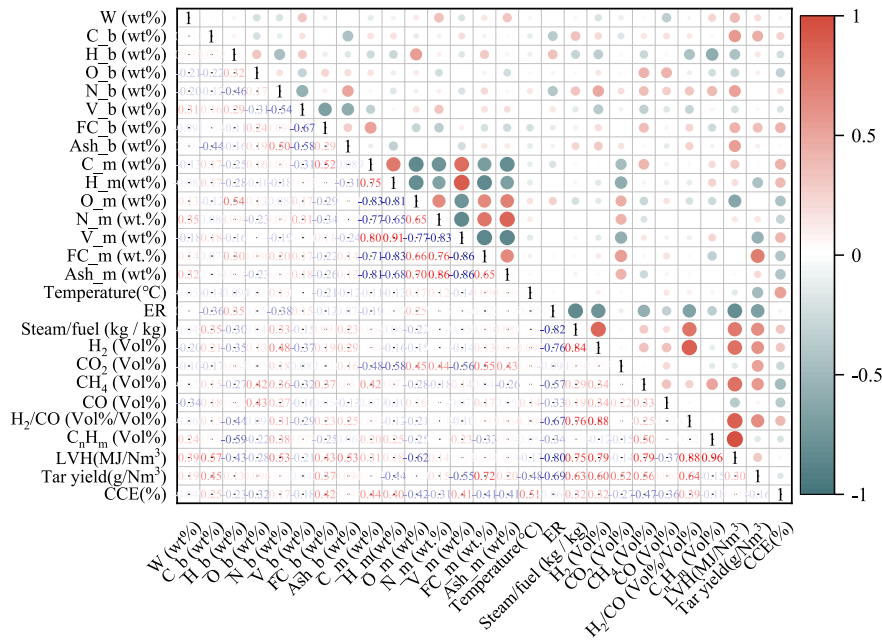


Fig. 3. Spearman correlation matrix between inputs and outputs.

mainly attributed to the high-molecular-weight polymer components in MSW, which increase the production of hydrocarbons and enhance the LHV syngas, but simultaneously increase tar content (0.19).

The PCA analysis for all features is shown in Fig. 4. It can be observed that for the three principal components, the first component mainly includes the ash content of biomass and the oxygen content of MSW, the second component includes the oxygen and hydrogen content of biomass, and the third component mainly includes the carbon and nitrogen content of biomass. This indicates the importance of the ultimate and proximate analyses of the feedstock in determining the three principal components. The proportions of variance explained by the three components are 35.4 %, 17.5 %, and 13.3 %, respectively, with a total of 66.2 %, indicating that the first three principal components capture a significant portion of the input characteristic information. The relationships between the different features are shown in Fig. 4(b). Features with similar scores will be placed in the same direction (positive importance), while features with different scores will be placed in opposite directions (negative importance). The fixed carbon content in biomass exhibits a good positive correlation with the volume fraction of CO. This can be attributed to the combustion reaction of oxygen and carbon and the reduction reaction of CO<sub>2</sub> with carbon, which promote the generation of CO from the fixed carbon. On the other hand, ER shows an inverse relationship with H<sub>2</sub> and CH<sub>4</sub> but a similar direction with CO<sub>2</sub>. This indicates the importance of ER control in co-gasification processes. A higher ER ensures a more complete combustion reaction, resulting in more carbon transforming into CO<sub>2</sub> and a lower concentration of combustible gases. Literature suggests that the optimal ER in gasification processes is within the range of 0.2–0.3 [2].

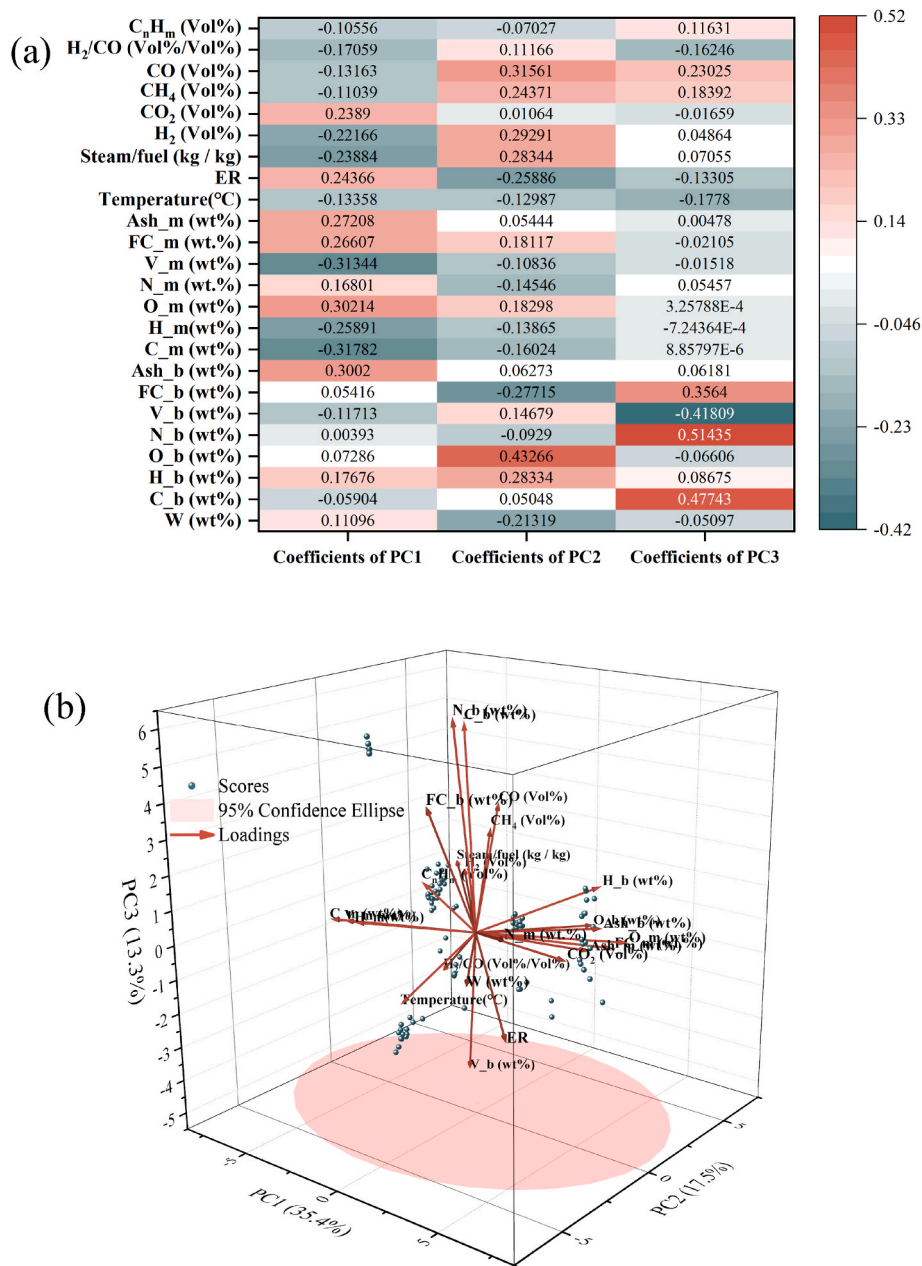
Fig. 5 demonstrates the impact of key operating parameters (temperature, ER, and steam/fuel) on the major syngas compositions (H<sub>2</sub>, CO, CH<sub>4</sub>, and CO<sub>2</sub>). From the figure, it can be observed that Steam/fuel has a significant effect on hydrogen production. Under high steam/fuel conditions, the proportion of H<sub>2</sub> in the synthesis gas can exceed 50 %. When the goal is to achieve a high H<sub>2</sub> concentration in gasification, the addition of steam for water-gas shift reactions becomes necessary. Within the gasification temperature range of 450–800 °C and steam/fuel ratio of 1.2–2.4, the volume fraction of hydrogen is relatively high. Additionally, the inclusion of steam seems to promote CO production,

mainly due to the forward shift of the water-gas reaction and methane reforming. When steam/fuel is above 0.4, and temperature ranges from 800–900 °C, with an ER ratio of 0.1–0.2, a higher CO volume fraction can be achieved.

### 3.2. Modeling results

In Table 2, quantitative statistics for the predictive performance of four ML models are presented, including the performance parameters R<sup>2</sup>, RMSE, and MAE. A higher R<sup>2</sup> value closer to 1, and lower RMSE and MAE values closer to 0, indicate better predictive performance of the models. It is evident from the table that all four models demonstrate good predictive performance for CO and H<sub>2</sub>, with R<sup>2</sup> values exceeding 0.9 in both the training and testing stage. This suggests that a wide distribution of data effectively enhances the model's training performance. The minimum RMSE values for CO and H<sub>2</sub> are 1.2 from HGBR and 1.6 from GBR, respectively, indicating minimal errors. For the prediction of CO<sub>2</sub> and CH<sub>4</sub>, it appears that the predictive accuracy is slightly lower than for CO and H<sub>2</sub>, but HGBR also demonstrates good predictive performance for both gases, with R<sup>2</sup> values of 0.93 and 0.88 in the testing stage, respectively. In the case of LHV prediction, RFR exhibits strong predictive performance with R<sup>2</sup> values of 0.98 in the training stage and 0.97 in the testing stage, along with the lowest RMSE value of 0.55. This suggests that the bagging approach used by RFR performs well and does not necessarily lag behind boosting methods, as their performance may vary depending on the dataset and the specific problem. It is worth noting that there seems to be some degree of overfitting for tar yield prediction, which may be related to the quantity of collected tar data.

Fig. 6 and Fig. S1 presents the actual values and predicted values for gasification performance parameters by the four ML models. In the figure, blue dots represent the training stage, red dots represent the testing stage, and the dashed line in the middle represents perfect alignment between actual and predicted values. Points clustered around the dashed line indicate good predictive results. It can be seen from the figure that the data points for CO, H<sub>2</sub>, CO<sub>2</sub>, CH<sub>4</sub>, H<sub>2</sub>/CO, C<sub>n</sub>H<sub>m</sub>, and LHV predictions are evenly distributed around the dashed line, indicating the applicability of the four models to the complex nonlinear



**Fig. 4.** PCA analysis of co-gasification. (a) correlation coefficients affecting the first three principal components and (b) the impact of parameter variable on syngas compositions in the information space.

thermochemical processes of biomass and MSW co-gasification. This ability to handle complex nonlinear problems is attributed to ensemble techniques, which do not rely solely on individual models but improve accuracy and reliability by combining multiple models. They can combine multiple weak learners (high bias, low variance) into a strong learner (low bias, high variance), thus enhancing predictive performance. By integrating the predictions of multiple models, errors from individual models can be reduced, resulting in a more accurate fit to the data [35]. This overcomes the limitations of individual models and improves predictive performance and generalization. On the other hand, it also reflects the importance of having suitable data for building ML models, as effective data is a crucial aspect of constructing excellent models.

Table 3 provides the optimal hyperparameters for each model. As all four machine learning models are derived from tree models, they share similar hyperparameters. By referring to the hyperparameter settings in

the table, it is possible to replicate the training of the machine learning models described in this paper. This, in turn, enables the achievement of the objective of predicting key gasification parameters for the co-gasification process of biomass and MSW in industrial applications using machine learning models. Overall, except for partial overfitting observed in the prediction of tar yield and CCE, the remaining parameters exhibit the optimal hyperparameter combinations, offering a comprehensive understanding for configuring each model.

### 3.3. Feature importance analysis

Feature importance analysis is a crucial step in the ML process. It involves evaluating the contribution of individual input features within a ML model to determine their impact on the model's prediction results. This helps us understand the importance of each feature, facilitating the elucidation of their interrelationships. In this study, we employed SHAP

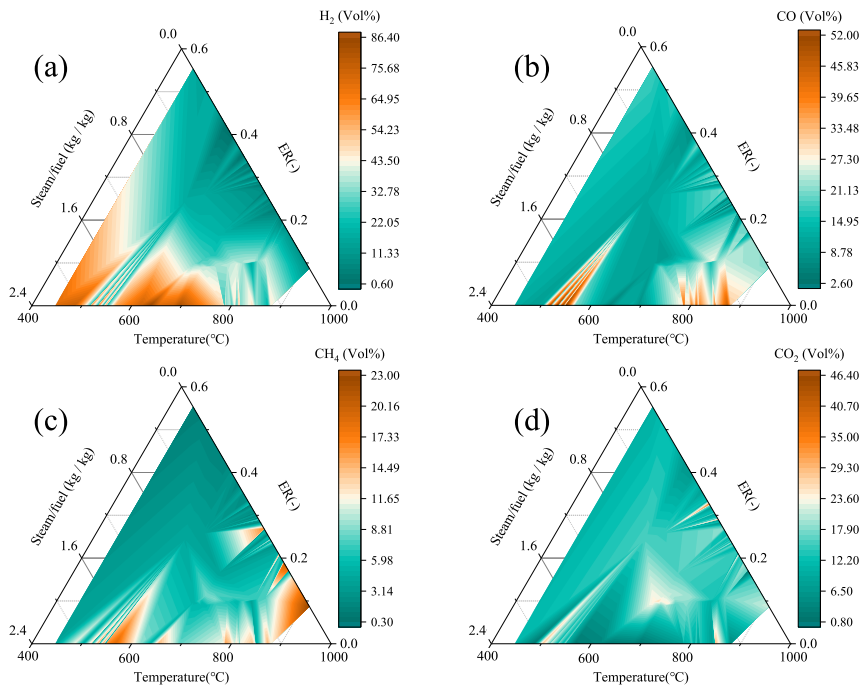


Fig. 5. The influence of operating parameters on the composition yield of syngas. (a) H<sub>2</sub>, (b) CO, (c) CH<sub>4</sub>, (d) CO<sub>2</sub>.

Table 2  
Statistical evaluation metrics for four ML models.

| Statistical parameter | Model type | CO (vol %) | H <sub>2</sub> (vol %) | CO <sub>2</sub> (vol %) | CH <sub>4</sub> (vol %) | C <sub>n</sub> H <sub>m</sub> (vol %) | H <sub>2</sub> /CO (vol%/vol %) | LHV (MJ/ Nm <sup>3</sup> ) | CCE (%) | Tar yield (g/ Nm <sup>3</sup> ) |
|-----------------------|------------|------------|------------------------|-------------------------|-------------------------|---------------------------------------|---------------------------------|----------------------------|---------|---------------------------------|
| <i>Training</i>       |            |            |                        |                         |                         |                                       |                                 |                            |         |                                 |
| R <sup>2</sup>        | XGBR       | 0.98       | 0.95                   | 0.89                    | 0.94                    | 0.87                                  | 0.99                            | 0.99                       | 0.98    | 0.71                            |
|                       | GBR        | 0.98       | 0.99                   | 0.87                    | 0.92                    | 0.84                                  | 1.0                             | 0.99                       | 0.98    | 0.91                            |
|                       | RFR        | 0.95       | 0.99                   | 0.86                    | 0.92                    | 0.85                                  | 0.99                            | 0.98                       | 0.98    | 0.94                            |
|                       | HGBR       | 0.98       | 0.98                   | 0.88                    | 0.94                    | 0.62                                  | 0.97                            | 0.96                       | 0.45    | 0.61                            |
| RMSE                  | XGBR       | 1.38       | 5.03                   | 2.42                    | 1.2                     | 0.85                                  | 0.18                            | 0.42                       | 2.5     | 6.29                            |
|                       | GBR        | 1.37       | 1.6                    | 2.64                    | 1.43                    | 0.97                                  | 0.12                            | 0.27                       | 2.24    | 0.18                            |
|                       | RFR        | 1.99       | 2.69                   | 2.76                    | 1.44                    | 0.93                                  | 0.27                            | 0.45                       | 2.52    | 2.82                            |
|                       | HGBR       | 1.2        | 2.93                   | 2.53                    | 1.22                    | 1.47                                  | 0.38                            | 0.75                       | 13.1    | 7.35                            |
| MAE                   | XGBR       | 0.94       | 2.74                   | 1.2                     | 0.83                    | 0.61                                  | 0.1                             | 0.24                       | 1.47    | 3.59                            |
|                       | GBR        | 0.94       | 1.03                   | 1.74                    | 1.01                    | 0.7                                   | 0.08                            | 0.09                       | 1.04    | 1.56                            |
|                       | RFR        | 1.44       | 1.28                   | 1.76                    | 0.95                    | 0.68                                  | 0.14                            | 0.26                       | 1.67    | 0.11                            |
|                       | HGBR       | 0.63       | 1.64                   | 1.51                    | 0.83                    | 1.06                                  | 0.17                            | 0.52                       | 10.4    | 5.17                            |
| <i>Testing</i>        |            |            |                        |                         |                         |                                       |                                 |                            |         |                                 |
| R <sup>2</sup>        | XGBR       | 0.94       | 0.94                   | 0.87                    | 0.86                    | 0.73                                  | 0.98                            | 0.97                       | 0.85    | 0.45                            |
|                       | GBR        | 0.92       | 0.97                   | 0.88                    | 0.85                    | 0.69                                  | 0.92                            | 0.94                       | 0.81    | 0.31                            |
|                       | RFR        | 0.91       | 0.97                   | 0.88                    | 0.88                    | 0.69                                  | 0.97                            | 0.97                       | 0.83    | 0.38                            |
|                       | HGBR       | 0.94       | 0.96                   | 0.93                    | 0.88                    | 0.77                                  | 0.97                            | 0.91                       | 0.01    | 0.2                             |
| RMSE                  | XGBR       | 2.25       | 5.62                   | 2.56                    | 1.35                    | 0.97                                  | 0.2                             | 0.56                       | 4.77    | 12.99                           |
|                       | GBR        | 2.53       | 3.72                   | 2.47                    | 1.39                    | 1.04                                  | 0.22                            | 0.78                       | 5.4     | 15.38                           |
|                       | RFR        | 2.66       | 3.86                   | 2.46                    | 1.25                    | 1.04                                  | 0.27                            | 0.55                       | 5.09    | 16.2                            |
|                       | HGBR       | 2.14       | 4.21                   | 1.85                    | 1.28                    | 0.88                                  | 0.27                            | 0.99                       | 12.8    | 17.12                           |
| MAE                   | XGBR       | 1.61       | 3.39                   | 1.7                     | 1.03                    | 0.62                                  | 0.14                            | 0.33                       | 3.47    | 7.84                            |
|                       | GBR        | 1.71       | 2.39                   | 1.85                    | 1.03                    | 0.81                                  | 0.16                            | 0.42                       | 3.8     | 7.9                             |
|                       | RFR        | 2.04       | 2.43                   | 1.92                    | 0.97                    | 0.79                                  | 0.18                            | 0.36                       | 3.75    | 8.62                            |
|                       | HGBR       | 1.52       | 2.79                   | 1.42                    | 0.97                    | 0.69                                  | 0.2                             | 0.68                       | 10.6    | 11.32                           |

analysis to assess the importance of input features on various output responses. This aids in comprehending the high-performance black-box ML model used in our research.

Fig. 7 and Fig. S2 presents the ranking of input feature importance that affects the performance parameters of biomass and MSW co-gasification. The importance of input features is arranged based on the average SHAP values, where larger average values indicate greater importance in influencing output responses. This is particularly valuable

in real industrial processes where increased attention should be given to highly influential features. In the right-hand graph, red points represent larger values of input features, while blue points indicate smaller values. The X-axis represents the magnitude of SHAP values. If the color of input feature values transitions from blue to red along the direction of increasing SHAP values, it indicates a positive influence of that input feature on the corresponding output response, and vice versa.

The factors that have the greatest impact on the volume fraction of

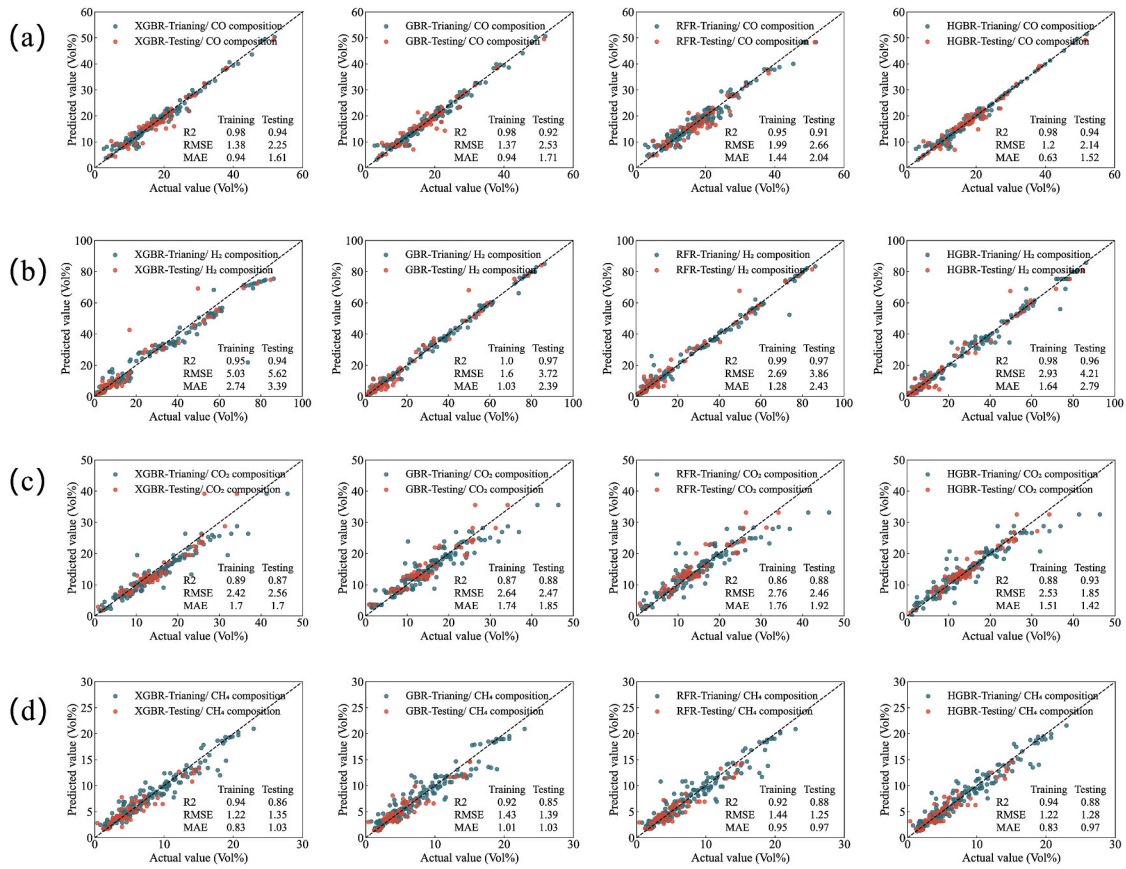


Fig. 6. Plot of predicted data against actual values during training and testing stage for four ML models. (a) CO, (b) H<sub>2</sub>, (c) CO<sub>2</sub>, (d) CH<sub>4</sub>.

Table 3

Optimal hyperparameters for each ML models.

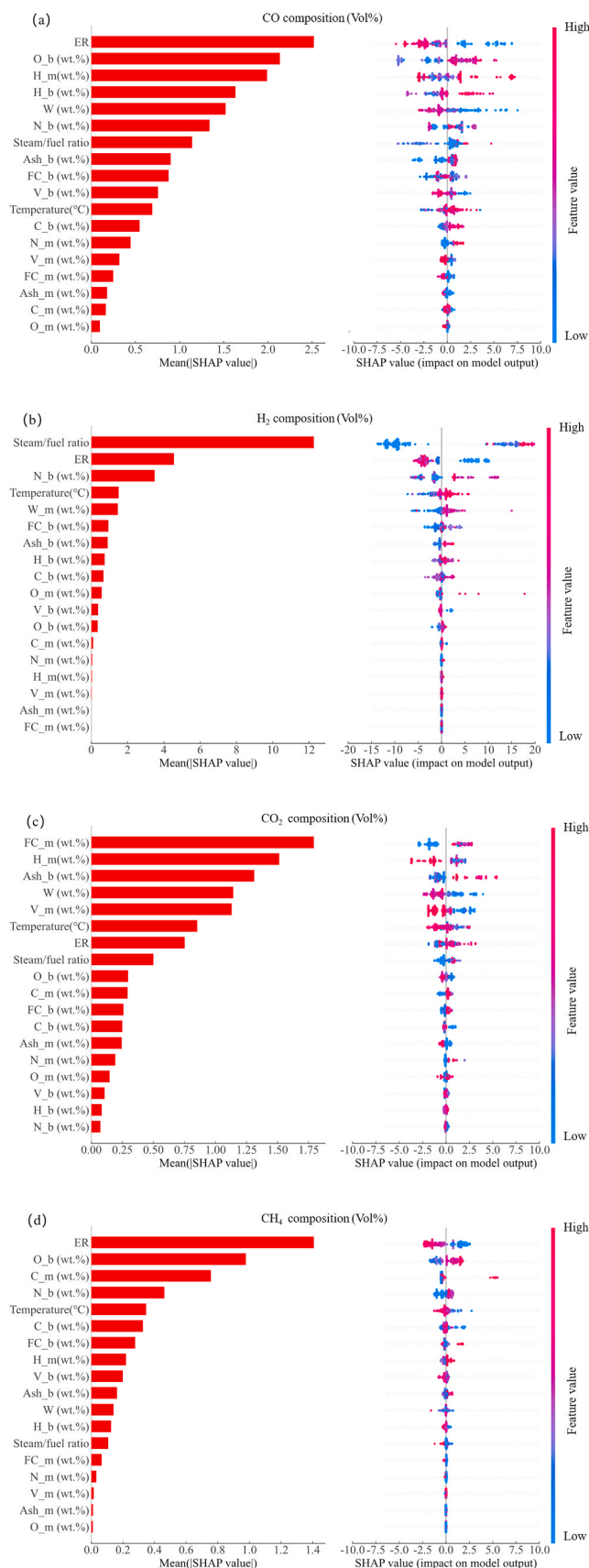
| Model | Hyperparameters   | CO (vol %) | H <sub>2</sub> (vol%) | CO <sub>2</sub> (vol %) | CH <sub>4</sub> (vol %) | C <sub>n</sub> H <sub>m</sub> (vol %) | H <sub>2</sub> /CO (vol%/vol %) | LHV (MJ/ Nm <sup>3</sup> ) | CCE (%) | Tar yield (g/ Nm <sup>3</sup> ) |
|-------|-------------------|------------|-----------------------|-------------------------|-------------------------|---------------------------------------|---------------------------------|----------------------------|---------|---------------------------------|
| XGBR  | n_estimators      | 79         | 21                    | 14                      | 98                      | 18                                    | 63                              | 71                         | 47      | 15                              |
|       | max_depth         | 29         | 32                    | 20                      | 21                      | 39                                    | 15                              | 28                         | 23      | 16                              |
|       | learning_rate     | 0.89       | 0.94                  | 0.4                     | 0.99                    | 0.92                                  | 0.98                            | 0.95                       | 0.94    | 0.95                            |
|       | subsample         | 0.68       | 0.37                  | 0.62                    | 0.46                    | 0.61                                  | 0.82                            | 0.69                       | 0.4     | 0.37                            |
| GBR   | n_estimators      | 175        | 76                    | 112                     | 197                     | 112                                   | 149                             | 200                        | 195     | 200                             |
|       | min_samples_split | 6          | 2                     | 9                       | 10                      | 6                                     | 6                               | 3                          | 3       | 3                               |
|       | learning_rate     | 0.39       | 0.32                  | 0.05                    | 0.05                    | 0.07                                  | 0.37                            | 0.17                       | 0.3     | 0.25                            |
|       | max_features      | 1          | 3                     | 4                       | 2                       | 2                                     | 4                               | 4                          | 2       | 1                               |
| RFR   | n_estimators      | 14         | 86                    | 58                      | 15                      | 109                                   | 170                             | 36                         | 134     | 23                              |
|       | min_samples_split | 5          | 2                     | 9                       | 5                       | 3                                     | 2                               | 2                          | 2       | 2                               |
|       | max_features      | 5          | 1                     | 5                       | 5                       | 5                                     | 5                               | 3                          | 1       | 2                               |
|       | max_depth         | 13         | 11                    | 11                      | 8                       | 7                                     | 13                              | 11                         | 12      | 8                               |
| HGBR  | max_iter          | 422        | 87                    | 218                     | 46                      | 10                                    | 432                             | 249                        | 212     | 12                              |
|       | max_leaf_nodes    | 92         | 40                    | 34                      | 39                      | 16                                    | 87                              | 51                         | 71      | 5                               |
|       | learning_rate     | 0.51       | 0.86                  | 0.05                    | 0.33                    | 0.94                                  | 0.1                             | 0.56                       | 0.13    | 0.96                            |
|       | max_bins          | 163        | 4                     | 200                     | 186                     | 51                                    | 87                              | 14                         | 227     | 133                             |
|       | max_depth         | 12         | 6                     | 13                      | 6                       | 12                                    | 13                              | 5                          | 6       | 6                               |

CO are ER, the oxygen content in biomass, the hydrogen content in biomass and MSW, and the blending ratio of MSW, with average SHAP values of 2.4, 2.1, 1.9, 1.7, and 1.6, respectively. A higher ER leads to negative shifts in SHAP values, indicating that a higher ER suppresses the generation of CO. This is reasonable to some extent as CO generation primarily arises from the primary pyrolysis of organic materials and the Boudouard reaction. Combustion simultaneously consumes CO, producing CO<sub>2</sub>. Excess O<sub>2</sub> accelerates the combustion reaction, leading to the depletion of CO.

Fig. 7(b) illustrates that the key features affecting H<sub>2</sub> concentration are the steam/fuel ratio and ER, with average SHAP values of 12.6 and

4.6, respectively. Steam-assisted gasification for hydrogen production is a current hot topic in the gasification field. Song et al. [9] have shown that the concentration of hydrogen in syngas is between 5 % and 25 % when air is used as the gasification agent, whereas with steam gasification, it ranges from 20 % to 60 %. This significantly enhances H<sub>2</sub> content. Steam gasification not only increases H<sub>2</sub> concentration but also greatly improves the H<sub>2</sub> to CO ratio, as indicated in Fig. S2(b). This positive effect is mainly attributed to the water gas shift reaction and steam methane reforming reaction [36].

The most critical feature influencing CO<sub>2</sub> concentration is the fixed carbon content in MSW. Carbon elements in fixed carbon are crucial for



**Fig. 7.** SHAP value obtained from the best-performing model representing the impact of input feature on each output target in biomass and MSW co-gasification. (a) CO, (b) H<sub>2</sub>, (c) CO<sub>2</sub>, (d) CH<sub>4</sub>.

CO<sub>2</sub> formation, and increasing the fixed carbon content drives the combustion reaction towards CO<sub>2</sub> generation. The generation of CO<sub>2</sub> reduces the heating value of syngas. Therefore, controlling the fixed carbon content in the mixed feedstock can also promote the production of high-quality syngas.

Syngas also contains low concentrations of hydrocarbons, primarily originating from the decomposition of volatile components. The hydrogen content and temperature in biomass are the key features influencing the concentration of this component, with average SHAP values of 1.1 and 0.82, respectively. In summary, the control of syngas composition relies significantly on the elemental composition of the feedstock and operating conditions such as ER and steam-to-feedstock ratio. Adjustments should be made based on the desired end products in the gasification process.

Figs. S2(c–e) illustrates the importance of input features on three other gasification parameters: LHV, CCE and tar yield. In terms of LHV, the most crucial influencing parameters are ER, the carbon content in biomass, and steam to feedstock ratio. A higher ER implies more N<sub>2</sub> entering the syngas, significantly reducing the LHV value. The addition of steam has a positive effect on increasing the heating value because it generates high-calorific-value H<sub>2</sub> while reducing the inclusion of N<sub>2</sub>. The increase in gasification temperature positively affects the CCE value, with an average SHAP value of 9.2, which aligns with the research of Inayat et al. [37]. The most important features affecting Tar yield are the Steam/fuel ratio, ER, the volatile content of MSW, and temperature, with average SHAP values of 3.8, 2.52, and 1.25, respectively. It's worth noting that the impact of the mentioned input features on result importance is based on the currently collected valid data. This conclusion may change as additional features are incorporated.

#### 4. Conclusions

This study conducted a detailed investigation of the co-pyrolysis process of biomass and MSW using ML models. Four state-of-the-art tree-based ML models (XGBR, GBR, RFR, and HGBR) were employed to model the co-gasification process of biomass and MSW. Results show that all four models exhibited excellent predictive performance, with R<sup>2</sup> values exceeding 0.9 in both the training and testing phases. HGBR exhibited the lowest RMSE values for predicting CO and GBR for predicting H<sub>2</sub>, with values of 1.2 and 1.6, respectively. The predictive performance of the four models for CO<sub>2</sub> and CH<sub>4</sub> was slightly lower than that for CO and H<sub>2</sub>. RFR showed good predictive performance for LHV, with R<sup>2</sup> values of 0.98 in the training phase and 0.97 in the testing phase, and the smallest RMSE value of 0.55. SHAP analysis was used to study the importance of input features on output responses. The most influential input features for CO concentration were ER, oxygen content in biomass, hydrogen content in biomass and MSW, and the blending ratio of MSW. The key features affecting H<sub>2</sub> concentration were Steam/fuel and ER.

Using data-driven ML models to study the co-gasification process of biomass and MSW allows for the rapid assessment of the impact of raw feedstock characteristics (proximate and ultimate analysis), operating conditions (temperature, gasifying agent, MSW blending ratio), and other factors on co-gasification performance parameters (syngas composition, LHV, tar yield). This offers significant advantages in terms of time and resource efficiency for industrial applications. However, it's essential to acknowledge the complexity of the co-gasification process. Some gasification methods, such as plasma gasification and hydrothermal gasification, were not considered in this study. Additionally, the role of catalysts was not modeled in this research. These aspects might be considered in subsequent studies.

#### CRedit authorship contribution statement

**Jingwei Qi:** Conceptualization, Data curation, Formal analysis, Investigation, Validation, Writing – original draft. **Yijie Wang:** Data

curation, Formal analysis. **Pengcheng Xu:** Formal analysis. **Ming Hu:** Methodology. **Taoli Huhe:** Supervision. **Xiang Ling:** Supervision. **Haoran Yuan:** Methodology, Supervision. **Yong Chen:** Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2023.130178>.

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